

Exploring Accessibility of Voice-Controlled Interfaces and Gesture Interactions in Vehicles for Secondary Driving Tasks for Deaf and Hard-of-Hearing Drivers

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Abstract

Voice Control Interfaces (VCIs) have become an ubiquitous technology, including in interacting with in-vehicle infotainment systems. However, research has revealed unique challenges VCIs pose for Deaf and hard of hearing (DHH) users due to reliance on voice as input and output. Despite extensive research on VCIs in home settings, there's a gap in understanding the specific challenges of VCIs in vehicles. To address this, we conducted a survey with 56 DHH participants, followed by interviews with 8 participants to explore their experiences and preferences regarding alternative interaction methods for in-vehicle VCIs focusing on secondary driving tasks. Results reveal technical, linguistic, and accessibility challenges affecting VCI efficiency in vehicles for DHH users, highlighting the need for alternative approaches. A follow-up brainstorming session with DHH participants suggests using gesture-based input and one-handed sign language as viable alternatives for in-vehicle VCIs, and exploring feasible gesture and sign language datasets for interaction with the in-vehicle systems.

CCS Concepts

• **Human-centered computing** → **Empirical studies in accessibility**.

Keywords

voice control interfaces, infotainment system, Deaf and Hard of Hearing, gesture control, accessibility, driving, vehicles

ACM Reference Format:

Sanskriti Kumar, Wendy Dannels, Tae Oh, and Roshan L Peiris. 2025. Exploring Accessibility of Voice-Controlled Interfaces and Gesture Interactions in Vehicles for Secondary Driving Tasks for Deaf and Hard-of-Hearing Drivers.

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719843>

In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3706599.3719843>

1 Introduction

Voice Control Interfaces (VCIs) have become ubiquitous across a range of technologies, including digital assistants like Google Home, Alexa, and Siri. However, they pose significant challenges for deaf and hard of hearing (DHH) users, primarily due to their reliance on speech for input and output [19]. These challenges are further enhanced in in-vehicle systems, where VCIs are increasingly adopted to enhance usability and safety [3, 5, 23]. Despite the potential advantages of hands-free, eyes-free interaction, VCIs often suffer from poor speech recognition accuracy, slow response times, and the need for users to remember specific commands [6, 18], making them inaccessible to DHH drivers. Additionally, current automatic speech recognition (ASR) technologies exhibit high error rates for Deaf speech due to insufficient and unrepresentative training data [7].

Research has explored alternative interaction methods, such as gestures for input and visual interfaces for feedback [9, 13], yet these studies largely overlook the specific needs of DHH drivers in in-vehicle contexts. Several studies have discussed communication and driving habits of DHH drivers [2, 10] that have identified unique issues, such as difficulties with phone calls and conversations with passengers while driving [12, 24], but they do not address the accessibility of voice or gesture-controlled systems in vehicles. Existing gesture-based systems often lack the accuracy, efficiency, and feedback mechanisms needed for real world usability, particularly for DHH users [15, 20].

To address these gaps, this research investigates the accessibility and usability challenges of in-vehicle VCIs for DHH drivers. Additionally, we explore feasible alternatives sign language gestures for input as an alternative for VCIs inspired by current gesture sensing systems in vehicles [11]. Specifically, we explore the following research questions:

- **RQ1:** What are the experiences of DHH drivers interacting with voice controlled in-vehicle systems for secondary driving tasks?

- RQ2: What are the preferences of DHH drivers for feedback from in-vehicle systems?
- RQ3: How would a DHH driver interact with a gesture controlled in-vehicle system?

This work adopts a multi-method approach, surveys and interviews to address RQ1 and RQ2 and a gesture elicitation study [22] to explore RQ3. By examining these aspects, this research aims to identify practical solutions for enhancing accessibility and usability, paving the way for inclusive in-vehicle interaction systems that cater to DHH drivers.

2 Methods

The survey aimed at gaining a preliminary quantitative understanding of how DHH drivers use in-vehicle systems for secondary driving tasks and also to recruit potential participants for the follow up interview and brainstorming sessions. Follow up interviews were conducted to get insights on the experience during these interactions. Both the survey and interview questionnaire were approved by the institute's Institutional Review Board.

2.1 Survey Method.

The online survey was designed using the Qualtrics software and included 24 questions (single choice, multiple choice, Likert scale, open-ended) that took at most 10 minutes to complete. The questionnaire was distributed in online DHH groups, and flyers in the National Technical Institute for the Deaf to recruit DHH participants. The survey aimed to understand the DHH drivers' method of completing secondary driving tasks such as interacting with the infotainment system in their vehicles, potential issues and challenges they have faced (particulars if/when using VCIs), the experience of feedback from in-vehicle VCIs, and their opinions on alternative methods of car interactions. Participants who did not use VCIs were asked why they chose not to use them. As compensation for participating in the survey, all respondents who consented were included in a lottery for 20 dollars.

2.1.1 Survey Participants. The survey was posted online for approximately 2 months. The survey collected 60 responses of which 54 were considered (6 were removed due to incomplete answers). 54 responses were from DHH drivers who either drive daily or occasionally. 22 were Deaf, 4 were deaf, and 27 were hard of hearing. Based on Carol Padden and Tom Humphries [16], for the survey and interview, 'Deaf' refers to individuals who are culturally deaf and 'deaf' refers to the physical condition of hearing loss. The remaining 3 participants had poor and moderate hearing and suffered from tinnitus. 48.3% of drivers use spoken language and 45.1% use sign language to communicate. Of the participants who use sign language, 65.2% use sign language for all conversations, 26% communicate using sign language with other sign language users, and 8.7% use it on a per-need basis.

2.2 Interview Method.

The interviews were conducted individually that followed a semi-structured format and started with learning about how the participants interact with in-car systems such as the infotainment system

and the challenges they faced, if any. They were asked to elaborate on the issues faced while using voice-controlled functions for both input and output. Additionally, the participants inquired about their experience with the feedback of the system and changes in infotainment systems that would benefit them. In the interview, a discussion on their experiences and perceptions of gesture control as an alternative interaction method, their expectations, and concerns was conducted. Lastly in the follow-up brainstorming session, each participant was asked to come up with gestures that they would use to control music, calls, and navigation in their vehicles. Finishing the interview, the participants were asked to share their opinions on gesture control after the brainstorming session.

2.2.1 Interview Participants. At the end of the survey, participants were asked if they would like to participate in a follow-up interview. Out of the 54 participants from the survey responses, 30 agreed to a follow-up interview. 8 participants who have had previous experience using in-vehicle VCIs were recruited for the interview and follow-up brainstorming session. 2 Deaf, 2 deaf, and 4 hard of hearing individuals were recruited for the interviews.

2.3 Data Analysis

The quantitative data from the valid responses from the survey were analyzed using suitable parametric and non-parametric methods. Next, the open-ended questions and the transcripts of the interviews were analyzed for themes in the participant feedback using Braun and Clarke's thematic analysis approach [4]. Due to the preliminary nature of the study and the repetitive themes identified in both survey and interview studies, we present the findings of both studies collectively. Here, note that the P1-P8 participated in both survey and interview studies (they were recruited from the survey) while the rest of the participants only participated in the survey study.

3 Findings

3.1 Current Usage Patterns and Experience with In-vehicle Systems

3.1.1 VCIs usage and interest. From the survey findings, it was identified that only 32% use or have tried using VCIs before. Out of those, 63.6% connect their smartphones to access the voice controlled personal assistants and 36.4% use the vehicle inbuilt VCIs. To wake up the VCIs, 52.5% use physical buttons on the steering wheel or the touch screen of their infotainment systems while 47.5% use voice commands. 61.1% of all participants stated that they do not use VCIs during their drive at all. It includes all of the Deaf participants, 66.7% of deaf participants, and 58.3% of hard of hearing participants. Among the interview participants, 2 participants use VCIs in vehicles frequently and 6 participants do not use VCI. Participants shared their interest in using VCIs but the system did not work the way they expected it to. P1 shared they have seen other drivers use it - "I want to use it. I would like to tell the car, play the next song, or call my mom, call my dad. I would like them to recognize it, but I don't know how to use it." Of those who used VCIs in vehicles, only 1 participant found it to be less distracting to the driving task. Some participants shared that typically, using physical buttons on the steering wheel is easy after they remember the functionality of each button.

3.1.2 VCI purposes. The primary tasks performed with the VCIs included accessing the infotainment systems in their car for three major functions: making calls (36.1%); inputting addresses for navigation (30.5%); and music (22.2%). The other functions included checking weather information and other tasks such as sending and reading texts, etc. Based on the survey, 44.9% of the drivers' primarily use physical buttons to interact with the infotainment system, followed by the touchscreen display (37.7%), and 10.2% use VCI.

3.1.3 Preferences towards the use of touchscreen and/or buttons. While participants had experience with VCIs, some participants preferred the use of tangible interfaces for interacting with the infotainment and in-car systems. In the interview P4 noted that they prefer the touchscreen because of its haptic feedback, *"...it gives me instant feedback when I touch it."* While the touch screen gives instant feedback, it may require drivers to move their eyes from the road and focus on the screen to interact. In contrast, participants noted that using buttons was accessible but not convenient for all functions. For example, P5 who found buttons to be helpful said *"I think it's easier with music because I know where all the buttons are for that. So I know what I'm pressing."* In contrast, P7 shared, *"The buttons are not descriptive sometimes, there are some options I still don't know what they are for."* and P1 shared *"I don't know what button to press on the steering wheel. There are too many buttons on the wheel."* Additionally, the display options and button options are far laid out concerning the driver's seat making it difficult for drivers to reach menu options while driving. Participants shared that it is often distracting and is a risk they take.

3.2 Challenges of using VCIs in Vehicles

Participants who used or have tried to use VCIs shared a common sentiment of wanting to use it for all functions but face different issues. From the survey, 75% of those who use VCI have faced issues while interacting with voice-controlled infotainment systems.

3.2.1 VCIs incapability to recognize DHH individuals' accents. 23 of the survey participants who did not use VCIs to access the in-vehicle systems, were deaf or Deaf and they either did not use spoken language or the system did not have the ability to comprehend deaf accent. Deaf participants relied on the touch screen and buttons and did not use VCIs. In the interview, P3 shared *"I don't use that because I don't ever speak with my voice. I only communicate in sign language."* and P5 shared that the system does not recognize their voice- *"I tried to use it once, it didn't interpret or read my voice. So I've got a deaf accent. It might sound a little different than a hearing user, but I tried it once. It didn't go so well, I don't use it."* Drivers with deaf accents had the same experience every time. P8: *"It doesn't recognize what I'm trying to say because I don't. I guess maybe I don't speak very well."* A hard of hearing participant with English as their second language had an added hurdle in using VCIs, P2: *"English is my second language. Even though I educated here (in the U.S.), but I didn't speak until age seven. So that's why maybe it's difficult."* The drivers either don't trust the system to understand their commands or blame their speech for the system's inability to understand. Two hard of hearing participants shared that the system had called the wrong person without confirming with the driver. This led to embarrassment and the driver feeling that they were at fault. The

confusion and distraction in such situations can be dangerous while the car is on the road.

3.2.2 Auditory feedback is inadequate for DHH drivers. Feedback to voice commands is crucial for user interaction with a VCI. The survey revealed that 46% of drivers find system feedback inadequate. P1 shared that they want to use VCI but are unable to understand its auditory response, *"Even if I use it (VCIs), I don't know what they say."* Participants appreciate confirmation of commands, but inconsistency can lead to confusion. 9 participants said that the system does not give them correct feedback to a command they give through VCIs. Issues arise especially when participants are unaware of command status, leading to distraction. Similarly, another participant drove to the wrong location due to the misinterpretation of the destination they had input using voice. They shared their frustration that the system did not auditorily confirm the exact address and executed it in the navigation application. Some participants face execution problems due to lack of confirmation or incorrect recognition, impacting their focus while driving.

3.3 Alternative accessible interactions to infotainment system

When the above-mentioned interaction methods don't work, participants resort to other ways like using their phone screens or asking their passengers to control the system. Sometimes, participants have to halt the vehicle and make the changes or execute the functions they need to which can waste their time. P1: *"I have to stop the car, do what I got to do (add location to navigation application), and then drive."*

3.3.1 Visually clear output is preferred. Participants expressed the need for visual responses and updates on the screen as they did not rely on auditory responses from the system. A deaf participant shared that they understand the safety risk involved in using a device while driving but that is the convenient interaction and interface for them, P8 shared, *"I'm trying to focus on driving, and I know we're not supposed to be on our devices while we're driving, so I try as hard as I can just to quickly glance over, select what I need to, and get my eyes back on the road."* Additionally, participants preferred a completely visual-based feedback that would include icons, clear text menus, subtitles for audio responses, and colored lights and haptic for emergencies. P3: *"My suggestion is to make everything visual, have it in text."* They preferred the dark mode to view the text as that was visually cleaner and faster to read. All participants emphasized they have better peripheral vision stating that they were comfortable with glancing at the screen while driving. Some suggested colors and noticeable changes for alerts, P5 suggested, *"I'll recognize in my peripheral vision that something's changed on the screen and that gives me the alert to look at it, but I don't always catch it. It would be nice if there was more colorful, more obvious, more something visually obvious that there was a change."* A hard of hearing participant shared that the option to control the pitch or tone of the auditory output would make VCIs accessible since they are able to hear specific pitches along with a visual display.

3.3.2 Transcription or subtitles for conversation and emergency alerts. Drivers preferred their text messages to be displayed on the screen, navigation options to be displayed on the screen to select

from, and in the case of audio output, responses to be transcribed and shown as subtitles on the screen. Hard of hearing drivers (P1 and P5) expressed that they wanted the system to transcribe spoken language conversations happening in the car so that they can be a part, as they currently feel left out when they are in the car as a driver or a passenger. 3 DHH drivers shared their concern about not being able to listen to emergency radio alerts or messages. They wanted the radio announcements to be either transcribed or displayed with lights on the screen. In case of emergency, 4 DHH drivers expressed that they wanted blinking and color-coded light alerts which would indicate situations of an emergency vehicle approaching, another car in the blind spot of the car, navigation alerts, or even text notifications. P6: *"When there's a siren, if there could be an alert that would interpret that sound so that we knew something was coming."*

3.4 Gesture or Sign Language as an Interaction method

Participants expressed willingness to explore gestures as an alternative interaction method to voice for secondary driving tasks. 54.9% of participants (19 hard of hearing, 14 Deaf, and all 4 deaf participants) preferred using gestures to control the infotainment systems in their vehicles.

3.4.1 Preferences to use sign language for interaction. Most drivers stated that using sign language and/or gestures will be an easier and faster interaction medium that would keep their eyes on the road. All interviewees felt that gestures would be helpful and were excited about using sign language as a natural language input as an alternative to VCIs or typing commands in their phones or vehicle displays. P8 shared *"It could just recognize my signing a lot faster and then maybe it would pull up the options. Probably quicker and easier than me trying to have to like type stuff in and do these things kind of have that physical interaction with the technology."* They suggested using one-handed signing and discussed it to be an intuitive interaction method. Signs in American Sign Language (ASL) were also suggested as a well-established dataset on which the system could be trained on for better recognition efficiency. Participants were interested in the possibility of signing and/or fingerspelling to enter addresses and destinations in navigation applications which will remove the need to stop the car or type information while driving.

3.4.2 Concerns about complex gestures, gesture/sign similarities and regional sign language differences. Participants emphasized that gestures should not be too complicated but not too general as well. Complicated gestures can be difficult for drivers to remember and general gestures can cause accidental recognition or execution of a function. 5 participants highlighted the need for confirmation after recognition of any gesture.

While gestures or sign language were discussed as easy and time-saving, participants had concerns that were similar to their VCIs experience. Misrecognition, misinterpretation, and accidental activation were issues they anticipated in the gesture control interaction despite its potential benefits.

Although they thought ASL would be an easy input method, some participants felt the regional differences in ASL and the sign

language differing internationally could pose challenges which is why the system should be trained to be able to recognize the differences. P5: *"sign language isn't universal. Each country or region has their own sign language, just like spoken languages. But gestures can be similar, like this gesture."* P8: *"instead of just having the system recognize one sign, if it could account for those regional dialects, too, just to provide more options, that would be great."*

3.5 Brainstorming gestures or signs for input

All interview participants were asked to participate in a brainstorming session where they were given commands and they had to come up with gestures or signs to control the car infotainment system. All participants used ASL in this study. Here, the participants discussed various aspects of using gestures/signs for interaction with in-vehicle systems such as the location for signing, wake up commands and signs for interaction.

3.5.1 Display and sensors closer to the driver makes gesture interaction easier. All participants in the interview study were right-handed signers. Thus, 5 participants preferred the camera or sensors to be on the right of the steering wheel and 3 participants preferred it on and in the center of the steering wheel to avoid moving their eyes. Participants also discussed the feasibility of integrating a sign detection confirmation (that the system is able to see the sign/gesture) to ensure that the command is being read.

3.5.2 Waking up the system and confirming input. To start their interactions, participants had to think of a gesture or sign they would use to wake up the system and signal it to start recognizing gestures as commands. Three participants suggested using an ASL sign for wake-up, and two preferred waving to the camera. Two participants preferred a physical button or a tap to activate the system. Beyond the wake-up commands, two participants wanted a confirmation when the system recognized a gesture before executing a command to avoid misrecognition.

3.5.3 Natural language interaction with ASL for commands. Participants emphasized on using one-handed signs for commands for safe driving. The participants were asked to command, 'play' and 'pause' the music along with increasing or decreasing the volume. Here, participants used ASL signs for 'play' using one hand (Fig. 1) and improvised one-handed sign of 'stop' to play/stop the music. When asked for possible signs for phone calls (Fig. 2), three participants used the ASL sign for 'pick', three signed 'phone' in ASL and others either signed 'yes' in ASL (1) or a thumbs up (1). Similar to which ASL sign for 'hang up' was suggested for ending a phone call. Another common suggestion was the idea to preset frequent locations like home or work addresses in the navigation application and access them using ASL (e.g. signing 'home') or signing numbers to their corresponding option. All participants preferred using ASL or fingerspelling the location or address they wanted to input. Similarly, the suggestion to sign 'gas' or 'food' would show the nearest gas station or restaurant respectively. 3 participants wanted to give commands as a hearing individual would do with a VCI i.e. sign the complete commands in ASL but with one hand.

3.5.4 Easy and intuitive gestures. Among gestures, to control music and command 'play', participants' ideas varied from waving a hand,



Figure 1: Participant using a combination of ASL and gestures to control music a: signing 'play' (shaking Y-handshape downward) with one hand to start playing the music b: palm facing gesture to command 'pause' the music.

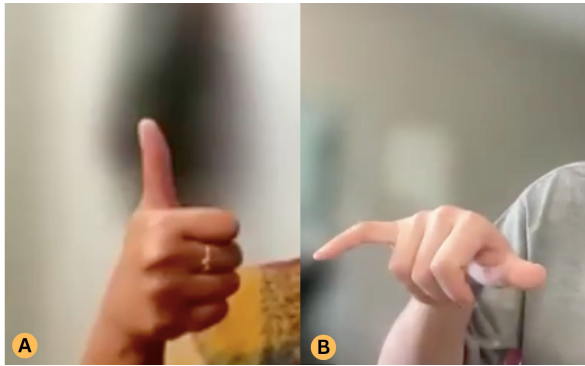


Figure 2: Participant using ASL to give commands for phone. a: sign for 'approval' to pick an incoming call b: sign for 'hang up' to end an ongoing call.

snapping fingers, tapping, or finger pointing, some participants agreed on a palm facing the camera to command 'pause' (Fig. 1). To control the volume, all participants suggested either a toggle of thumbs up and down or a hand up and down motion to increase or decrease the volume respectively. 2 participants came up with a slithering (like a snake) movement to indicate a route or a road. Lastly, some participants suggested finger-pointing, swipes, and pinching gestures near the steering wheel to avoid having to reach all the way to touch the screen.

4 Discussion

The study investigated the current use of and experience with voice control interfaces. In answering RQ1, a prominent theme that emerged was the interest of DHH drivers in using VCIs which was based on the ease of functionality it provides to them or the drivers around them. However, a majority of the DHH population does not use voice control due to the inaccessibility of the input method. Those who did make an effort to use VCIs, they often found themselves repeating commands because the system did not interpret deaf accents resulting in frustration, distracted driving,

and finally resorting to other interaction methods. While some of these challenges have been observed in previous research on VCIs in home settings [8], we highlight the additional issues such as cognitive load while driving, communication with passengers, etc. in this research in the driving context. Hence, drivers were completing secondary driving tasks like using the infotainment system through conventional interaction methods of buttons or touch screens which also came with the issue of overwhelming number of options causing distraction while driving.

In answering RQ2, concerns regarding the lack of feedback for DHH individuals were highlighted in the interviews. Alternative feedback systems emerged as a crucial aspect [14], with participants discussing personalized modalities based on their needs. DHH drivers emphasized clear visual outputs in their range of vision while driving to minimize distractions (such as on heads-up displays[21]). Suggestions included subtitles for audio responses, the choice to change the pitch and tone of audio or transcription of spoken language in the vehicle, and color-coded light alerts for emergencies. Subtitles and a sign language interpretation to voice for phone calls have been suggested in previous studies as technologies for communication between professional drivers and passengers [12] but it surfaced as a need in the interviews for personal conversations among DHH drivers and their passengers.

While research has explored interest of drivers for using gesture while driving [11, 17, 20], in answering RQ3, signing as an input methods garnered significant interest among DHH participants as an accessible alternative to VCIs. Through the interview and follow-up brainstorming session, it was found that DHH drivers preferred socially used and toggle gestures for commands like increase/decrease and switching on/off functions. As the commands became specific like opening an application or adding an address, participants regardless of their hearing status preferred to use sign language to input commands or fingerspelling names and addresses. While sign language were perceived as a convenient alternative to voice, they present other challenges such as more nuanced input requires recognizing gaze and facial expressions [1]. Thus, concerns regarding recognition accuracy were noted, mirroring challenges experienced with VCIs and sign language recognition systems.

5 Limitations and Future Works

Our focus was on understanding the experience of DHH drivers with VCIs, and more participants in the interview would allow us to gain elaborate insights. Contextual inquiries and simulations to identify further accessibility challenges could inform improvements in the interaction between drivers and in-vehicle systems. Systematic research could explore improving voice recognition algorithms, which would involve training models on a broader dataset to better comprehend diverse accents, including those of DHH individuals and individuals who have English as their second language.

Accessibility and cognitive load of hand gestures: Our study did not evaluate the potential use of gestures or one-handed signs in actual driving scenarios. For a standard set of gestures, the cognitive load to remember and use them should be considered for diverse drivers who may have different cognitive, auditory, and/or physical needs.

Sign language recognition technology: Investigating the feasibility and effectiveness of sign language or gesture recognition technologies as alternative input methods needs further exploration. Future research could focus on developing efficient sign recognition systems for varying car environments, capable of accurately interpreting sign languages from different regions and countries.

6 Conclusion

In this research, we conducted a survey, interviews and follow-up brainstorming session with DHH drivers to understand the unique usability and accessibility characteristics of VCIs in vehicles. The study underlines the inaccessibility of VCIs and inefficiency of other interaction methods for DHH drivers. The highlight is the need and interest in an accessible alternative i.e. sign language for a natural interaction method for secondary driving tasks along with visual outputs. Further research is needed to develop recognition systems and the input gesture dataset considering the needs of different users.

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